

Problem 1 Given $f: X \rightarrow X$ and $i_X: X \rightarrow X$ the identity map.

(i) By definition of the composition of two maps

$f \circ i_X: X \rightarrow X$ and $i_X \circ f: X \rightarrow X$. Consequently the three maps in the statement have the same domain and range. They also have the same rule of assignment because

$$f \circ i_X(x) = f(i_X(x)) = f(x) \quad \forall x \in X$$

$$i_X \circ f(x) = i_X(f(x)) = f(x) \quad \forall x \in X$$

which establishes the equality.

(ii) Since in this case the domain and the range of

$f: X \rightarrow X$ coincide, so is the case for $f^{-1}: X \rightarrow X$

and we can write the two compositions $f \circ f^{-1}: X \rightarrow X$

and $f^{-1} \circ f: X \rightarrow X$.

Then, for all $x \in X$ and by definition of $f^{-1}: X \rightarrow X$

(2)

• $f \circ f^{-1}(x) = f(y)$ where $y \in X$ is such that

$$f(y) = x \Rightarrow f \circ f^{-1}(x) = x = i_x(x)$$

• $f^{-1} \circ f(x) = f^{-1}(f(x)) = z$ where $z \in X$ is

such that $f(z) = f(x)$. Since f is injective

$$\Rightarrow z = x \Rightarrow f^{-1} \circ f(x) = x = i_x(x).$$

(iii) let $g: X \rightarrow X$ such that $g \circ f = f \circ g = i_x$. Then

$$g \underset{\text{part (i)}}{=} g \circ i_x \underset{\text{part (ii)}}{=} g \circ (f \circ f^{-1}) = (g \circ f) \circ f^{-1} = i_x \circ f^{-1}$$

$$\underset{\text{part (i)}}{=} f^{-1}.$$

Problem 2

③

(i) We prove separately the two implications.

$\Rightarrow$) Suppose that f is one-to-one. Then if $y \in f(X)$ then there exists a unique point $x \in X$ such that $f(x) = y$, because if $f(x) = f(z) = y$ then $x = z$ because f is injective. Having this in mind, define

$$g : Y \rightarrow X \quad \text{by} \quad g(y) = \begin{cases} f^{-1}(y) & \text{if } y \in f(X) \\ a \in X & y \notin f(X) \end{cases}$$

Clearly

$$g \circ f(x) = g(f(x)) = f^{-1}(f(x)) = x \quad \Rightarrow$$

$$g \circ f = i_X.$$

$\Leftarrow$) Let $x, y \in X$ s.t. $f(x) = f(y)$. Then

$$g(f(x)) = g(f(y)) \quad \Rightarrow \quad x = y.$$

(ii) $\Rightarrow$) f onto $\Rightarrow f(X) = Y$ and (4)

hence, for any $y \in Y$ we can choose $x \in X$
and $h: Y \rightarrow X$ s.t. $h(y) = x$ (this implies
using the Axiom of Choice, more in Problem 6)

Consequently, for any $y \in Y$

$$f \circ h(y) = f(x) = y \Rightarrow f \circ h = i_Y$$

$\Leftarrow$) Given $y \in Y$, then $h(y) \in X$ is s.t.

$$f(h(y)) = y \Rightarrow f \text{ is onto. } \#$$

Problem 3 $\Rightarrow$) See Problem 1 part (ii)

$\Leftarrow$) Problem 2 (i) shows that f is injective and
part (ii) that f is surjective. and hence f is
a bijection. Problem 1 part (iii) proves that
there is only one such map.

(5)

Problem 4

(i) We first prove the inclusion $f(f^{-1}(B)) \subseteq B$

Let $y \in f(f^{-1}(B)) \Rightarrow \exists z \in f^{-1}(B)$ such that

$y = f(z)$. As $z \in f^{-1}(B) \Rightarrow \exists u \in B$ s.t.

$f(z) = u \in B$. This implies that

$$y = f(z) = u \in B \Rightarrow f(f^{-1}(B)) \subseteq B.$$

We now show the equivalence

$$f(f^{-1}(B)) = B \text{ for all } B \Leftrightarrow f \text{ is onto}$$

$$\Rightarrow) f(f^{-1}(B)) = B \text{ for all } B \subseteq Y$$

implies that $Y \subseteq f(f^{-1}(Y)) \subseteq Y$ and

hence $f(f^{-1}(Y)) = Y \Rightarrow f(X) = Y \Rightarrow f \text{ onto}$

$\Leftarrow)$ Let $B \subseteq Y$ and $b \in B$. f onto implies

$\exists x \in X$ such that $f(x) = b \Rightarrow x \in f^{-1}(B)$

and hence $b \in f(f^{-1}(B)) \quad \# \quad \textcircled{6}$

(i) We first prove the inclusion $A \subset f^{-1}f(A)$.

Let $a \in A \Rightarrow f(a) \in f(A) \Rightarrow a \in f^{-1}(f(A))$

We now show the equivalence

$A = f^{-1}f(A)$ for all $A \subset X \Leftrightarrow f$ is injective.

This time we use problem 2 part (i).

$\Rightarrow$) Taking as set A the singleton $\{x\} \subset X$

we have that $x = f^{-1}(f(\{x\})) \quad (*) \quad \forall x \in X$

Hence if $x, x' \in X$ are s.t. $f(x) = f(x')$

$(*)$ implies that $x = x'$ necessarily and hence f is injective.

Note: The f^{-1} in $(*)$ is the preimage not the inverse!!!

⑦

$\Leftarrow$) Let $A \subset X$ and $x \in f^{-1}(f(A))$

$\Rightarrow f(x) \in f(A)$ which implies that $\exists a \in A$

s.t. $f(x) = f(a)$. Since f is injective $\Rightarrow$

$\Rightarrow x = a \in A \Rightarrow A \subset f^{-1}(f(A))$ $\neq$

(iii) $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$ in
general (prove!)

We show $\Rightarrow$) Suppose there exists $x, x' \in X$, $x \neq x'$

and $f(x) = f(x')$. Then

$$f(x) = f(\{x\}) = f(\{x\}) \cap f(\{x'\}) =$$

$$f(\{x\} \cap \{x'\}) = f(\emptyset) \text{ which contradicts}$$

the hypothesis that f is a function. This implies

that $x' \neq x$ is not possible $\Rightarrow f$ injective.

($\Leftarrow$) let $A_1, A_2 \subset X$ and $y \in f(A_1) \cap f(A_2)$ ⑧

$\Rightarrow y = f(a_1) = f(a_2)$ with $a_1 \in A_1, a_2 \in A_2$

f injective $\Rightarrow a_1 = a_2 \Rightarrow a_1 = a_2 \in A_1 \cap A_2$

$\Rightarrow f(A_1) \cap f(A_2) \subseteq f(A_1 \cap A_2)$

~~$\nsubseteq$~~

(iv) $Y \setminus f(A) \subseteq f(X \setminus A)$ for all $A \subset X$

$\Rightarrow f$ is onto. Indeed, let $y \in Y \setminus f(X)$.

By hypothesis $y \in f(X \setminus X) = f(\emptyset) = \emptyset$

which implies that such a y does not exist

and hence $f(X) = Y$ necessarily

(conversely, suppose f is onto and let

$y \in Y \setminus f(A)$. Since $Y = f(X) \Rightarrow \exists x \in X$

s.t. $y = f(x)$ and $y \notin f(A) \Rightarrow$

$y \in f(X \setminus A)$. ~~$\nsubseteq$~~

Problem 5 In order to simplify the notation, for each 9
element $x \in X$, we will denote by $[x]$ the equivalence
class that contains it and determined by $\sim$.

With this notation we have to show that

i) The classes $[x]$ are non-empty

ii) $\bigcup_{x \in X} [x] = X$

iii) Given two classes $[x]$, $[y]$ we have
that either $[x] = [y]$ or $[x] \cap [y] = \emptyset$.

We go one by one:

i) $x \in [x]$ by the reflexivity of $\sim$ and
hence $[x]$ is non-empty

ii) Since by part (i) $x \in [x]$ we have
that $\{x\} \subset [x]$ and hence

$$X = \bigcup_{x \in X} \{x\} \subset \bigcup_{x \in X} [x] \subset X$$

By double inclusion $\bigcup_{x \in X} [x] = X$.

(iii) let $[x], [y]$ be s.t. $[x] \cap [y] \neq \emptyset$ (10)

let $z \in [x] \cap [y]$ and $u \in [x]$ arbitrary.

$$\begin{aligned} \text{Since } z \in [x] &\Rightarrow z \sim u \Rightarrow u \sim z \\ z \in [y] &\Rightarrow z \sim y \end{aligned}$$

By the transitive property $u \sim z$ & $z \sim y \Rightarrow u \sim y$
which implies that $u \in [y]$. Since $u \in [x]$
is arbitrary, we have shown that

$$[x] \subset [y]$$

Analogously, it can be shown that $[y] \subset [x]$
and hence $[x] = [y]$, as required. #

Solutions of Tutorial #1

Topics covered:

- Sets and functions
- Relations and cardinalities

Problem 6

(i) We show the three properties necessary for $\sim$ to be an equivalence relation:

Reflexivity: Consider any $a \in A$ so that of course $f(a) = f(a)$ since f is a function. Hence $a \sim a$ so that $\sim$ is reflexive since a was arbitrary.

Symmetry: Consider $a, b \in A$ and suppose that $a \sim b$. Then by definition $f(a) = f(b)$ so that obviously also $f(b) = f(a)$ since equality is symmetric. So of course $b \sim a$, which shows that $\sim$ is symmetric.

Transitivity: Consider $a, b, c \in A$ and suppose that $a \sim b$ and $b \sim c$. Then by definition $f(a) = f(b)$ and $f(b) = f(c)$ so that of course $f(a) = f(b) = f(c)$, and hence $a \sim c$. This shows that $\sim$ is transitive.

(ii) Proof. Define the function $g : A^* \rightarrow B$ as follows. For any equivalence class $C \in A^*$, we know that C is nonempty since A^* is a partition of A . Hence there is an $a \in C$, so set $g(C) = f(a)$, noting that clearly $g(C) = f(a) \in B$ so that B can be the range of g . To show that g is injective, consider two equivalence classes C and D where $g(C) = g(D)$. Then there are elements $c \in C$ and $d \in D$ where $f(c) = g(C) = g(D) = f(d)$. This shows that $c \sim d$ so that they must be in the same equivalence class. Thus $d \in C$ since $c \in C$, but also $d \in D$ so that C and D are not disjoint. Hence it must be that $C = D$ by Lemma 3.1, which shows that g is injective.

To show that g is surjective, consider any $b \in B$. Since f is surjective, there is an $a \in A$ such that $f(a) = b$. Since A^* is a partition, a must belong to an equivalence class $C \in A^*$. Then there is an element $c \in C$ such that $g(C) = f(c)$ by the definition of g . Since a and c are both in the same equivalence class C , we have that $a \sim c$ so that $g(C) = f(c) = f(a) = b$. This shows that g is surjective since $b \in B$ was arbitrary.

Therefore we have shown that g is both injective and surjective, and so is a bijection by definition, as desired.

Problem 7

Clearly $<$ is a relation on $A \times B$ so we just need to show the three properties of an order relation:

Comparability: Consider distinct points $a_1 \times b_1$ and $a_2 \times b_2$ in $A \times B$ so that $a_1 \neq a_2$ or $b_1 \neq b_2$. If $a_1 \neq a_2$ then they are comparable in $<_A$ (since it is an order relation) so that, without loss of generality, we can assume that $a_1 <_A a_2$. Then of course $a_1 \times b_1 < a_2 \times b_2$ by definition. So assume that $a_1 = a_2$ so that it must be that $b_1 \neq b_2$. Then b_1 and b_2 are comparable in $<_B$ since it is an order relation. So without loss of generality we can assume that $b_1 <_B b_2$. Then of course $a_1 \times b_1 < a_2 \times b_2$ since also $a_1 = a_2$. Thus either way $a_1 \times b_1$ and $a_2 \times b_2$ are comparable in $<$.

Nonreflexivity: Suppose that $a \times b$ is any element of $A \times B$. Since $<_B$ is an order relation, it is nonreflexive so that it is not true that $b <_B b$. Since of course $a = a$, it follows that it cannot be that $a \times b < a \times b$ since it would have to be that $b <_B b$. Hence $<$ is nonreflexive since $a \times b$ was arbitrary.

Transitivity: Suppose that $a_1 \times b_1 < a_2 \times b_2$ and $a_2 \times b_2 < a_3 \times b_3$. We then have the following cases:

- Case: $a_1 <_A a_2$. Still two cases:
 - Case: $a_2 <_A a_3$. Then $a_1 <_A a_2$ and $a_2 <_A a_3$ so that $a_1 <_A a_3$ since $<_A$ is transitive. Thus by definition $a_1 \times b_1 < a_3 \times b_3$.
 - Case: $a_2 = a_3$ and $b_2 <_B b_3$. Then $a_1 <_A a_2 = a_3$ so that $a_1 \times b_1 < a_3 \times b_3$ by definition.
- Case: $a_1 = a_2$ and $b_1 <_B b_2$. Still two cases:
 - Case: $a_2 <_A a_3$. Then $a_1 = a_2 <_A a_3$ so that by definition $a_1 \times b_1 < a_3 \times b_3$.
 - Case: $a_2 = a_3$ and $b_2 <_B b_3$. Then $a_1 = a_2 = a_3$ and $b_1 <_B b_2$ and $b_2 <_B b_3$ so that $b_1 <_B b_3$ since $<_B$ is transitive. Therefore again $a_1 \times b_1 < a_3 \times b_3$ by definition.

Thus in all cases $a_1 \times b_1 < a_3 \times b_3$, which shows that $<$ is transitive.

Problem 8

We define a function $f : A \times B \rightarrow B \times A$. For any element $(a, b) \in A \times B$ we set $f(a, b) = (b, a)$, noting that of course $a \in A$ and $b \in B$. It should be obvious then that $f(a, b) = (b, a) \in B \times A$ so that $B \times A$ can be the range of f .

First we show that f is injective. To this end consider (a_1, b_1) and (a_2, b_2) in $A \times B$ where $(a_1, b_1) \neq (a_2, b_2)$. Of course we have that $f(a_1, b_1) = (b_1, a_1)$ and $f(a_2, b_2) = (b_2, a_2)$. Since $(a_1, b_1) \neq (a_2, b_2)$ clearly either $a_1 \neq a_2$ or $b_1 \neq b_2$. In either case it should be clear that $f(a_1, b_1) = (b_1, a_1) \neq (b_2, a_2) = f(a_2, b_2)$, which shows that f is injective since (a_1, b_1) and

(a_2, b_2) were arbitrary. It is very easy to see that f is also surjective since, for any $(b, a) \in B \times A$, clearly $(a, b) \in A \times B$ and $f(a, b) = (b, a)$. Hence f is a bijection as desired. Note that if $A \times B = \emptyset$ then $f = \emptyset$ as well, which is vacuously a bijective function since it must be that $B \times A = \emptyset$ as well (because either $A = \emptyset$ or $B = \emptyset$).

Problem 9

Suppose that B is not finite and $B \subset A$ but that A is finite. Since $B \subset A$, either $B = A$ or B is a proper subset of A . In the former case we clearly have a contradiction since B would be finite since A is and $B = A$. In the latter case we have that there is a bijection from A to $\{1, \dots, n\}$ for some $n \in \mathbb{Z}_+$ by definition since A is finite. Then, since B is a proper subset of A , it follows from Theorem 6.2 that there is a bijection from B to $\{1, \dots, m\}$ for some $m < n$. However, then clearly B is finite by definition, which is also a contradiction since we know B is not finite. Hence in either case there is a contradiction so that A must not be finite.

Problem 10

As is customary, denote the set of all functions from A to B by B^A . First, if $A = \emptyset$, then the only function from A to B is the vacuous function $\emptyset$ so that $B^A = \{\emptyset\}$, which is clearly finite. So assume that $A \neq \emptyset$. Then, since A is finite, there is a bijection f from A to $\{1, \dots, n\}$ for some $n \in \mathbb{Z}_+$, noting that of course f^{-1} is then a bijection from $\{1, \dots, n\}$ to A .

We construct a bijection h from B^A to B^n . So, for any $g \in B^A$ set $h(g) = g \circ f^{-1}$, noting that clearly this is a function from $\{1, \dots, n\}$ to B . Hence h is a function from B^A to B^n .

To show that h is injective consider g and g' in B^A where $g \neq g'$. It then follows that there is an $a \in A$ where $g(a) \neq g'(a)$. Then let $k = f(a)$ so that clearly $f^{-1}(k) = a$ and $k \in \{1, \dots, n\}$. We then have that $(g \circ f^{-1})(k) = g(f^{-1}(k)) = g(a) \neq g'(a) = g'(f^{-1}(k)) = (g' \circ f^{-1})(k)$ so that it must be that $h(g) = g \circ f^{-1} \neq g' \circ f^{-1} = h(g')$. Since g and g' were arbitrary, this shows that h is injective.

Now consider any function $i \in B^n$ and let $g = i \circ f$ so that clearly g is a function from A to B since $f : A \rightarrow \{1, \dots, n\}$ and $i : \{1, \dots, n\} \rightarrow B$. Hence $g \in B^A$, and $h(g) = g \circ f^{-1} = (i \circ f) \circ f^{-1} = i \circ (f \circ f^{-1}) = i$. Since i was arbitrary, this shows that h is surjective as well.

Hence h is bijection from B^A to B^n . Now, since B^n is a finite cartesian product of finite sets (since B is finite), it is finite by Corollary 6.8. Thus it must be that B^A is also finite since there is bijection between them.